Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology



Course Code
ECE 3112

Course Title
Microprocessor, Assembly Language & Interfacing Sessional

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Lab Report 4: Using Assembly Language to Program 8086 Microprocessor to Solve Problems.

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Problem 4.1

Problem Description

Write a code to find the min/max number from three numbers (Using registers).

Tools Used

- emu8086 - 8086 Microprocessor Emulator
- MacTeX -LaTeXcompiler.
- VS Code with LaTeXworkshop extension as a text editor.

Process

Code:

```
—

DATA SEGMENT
    MSG1 DB "The max number is: "
    ANS DB ?, '$'
DATA ENDS

CODE SEGMENT
    ASSUME CS:CODE, DS DATA

START:
    MOV AX, DATA
    MOV DS, AX

    MOV AX, '8'
    MOV BX, '3'
    MOV CX, '6'

    CMP AX, BX
    JG CHK1

    MOV DX, BX
    JMP NXT1

CHK1:
    MOV DX, AX
```

```
25
26  NXT1:
27      CMP DX, CX
28      JG CHK2
29      MOV AX, CX
30      JMP NXT2
31
32  CHK2:
33      MOV AX, DX
34
35  NXT2:
36      MOV ANS, AL
37      LEA DX, MSG1
38      MOV AH, 9
39      INT 21H
40      MOV AH, 4CH
41      INT 21H
42
43  CODE ENDS
44  END START
```

Output

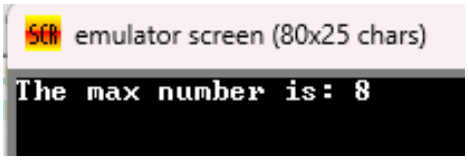


Figure 1: Finding min/max number

Problem 4.2

Problem Description

Write a program to display a "?", read two capital letters, and display them on the next line in alphabetical order.

Process

Code:

```
—

DATA SEGMENT
    CR EQU 0DH
    LF EQU 0AH
    MSG1 DB CR, LF, '? $'
    CHAR1 DB ?
    CHAR2 DB ?
DATA ENDS

CODE SEGMENT
    ASSUME CS:CODE, DS:DATA

START:
    LEA DX, MSG1
    MOV AH, 9
    INT 21H

    MOV AH, 1
    INT 21H
    MOV CHAR1, AL

    MOV AH, 1
    INT 21H
    MOV CHAR2, AL

    MOV AL, CHAR1
    MOV BL, CHAR2

    CMP AL, BL
    JLE SKIP_SWAP
    MOV CHAR1, BL
    MOV CHAR2, AL

SKIP_SWAP:
    ; newline
    MOV AH, 2
    MOV DL, 0Dh
    INT 21h
```

```
38      MOV DL, 0Ah
39      INT 21h
40
41      ; first character
42      MOV AH, 2
43      MOV DL, CHAR1
44      INT 21h
45
46      ; second character
47      MOV AH, 2
48      MOV DL, CHAR2
49      INT 21h
50
51      ; Terminate program
52      MOV AH, 4Ch
53      INT 21h
54
55  CODE ENDS
56
57  END START
```

Output

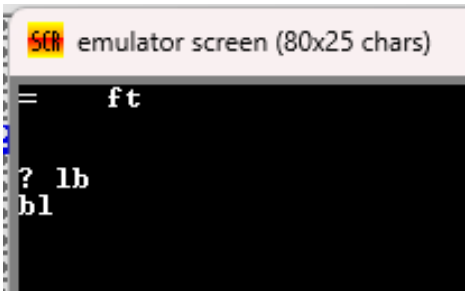


Figure 2: Ordering letters alphabetically

Problem 4.3

Problem Description

Write a program that will prompt the user to enter a HEX digit character(”0”,.....”9” or, ”A”,.....”F”), display it on the next line in decimal, and ask the user if he or she wants to do it again. If the user types ”Y” or ”y”, the program terminates. If the user enters an illegal character. prompt the user try again.

Process

Code:

```
—

DATA SEGMENT
    CR EQU 0DH
    LF EQU 0AH
    MSG1 DB CR, LF, "Enter A HEX Digit: $"
    MSG2 DB CR, LF, "Do it again? $"
    MSG3 DB CR, LF, "In Decimal: $"
    MSG4 DB CR, LF, "In Decimal: 1$"
    MSG5 DB CR, LF, "Try Again.$"
    CHAR DB ?, '$'
    ANS DB ?, '$'
DATA ENDS

CODE SEGMENT
    ASSUME CS:CODE, DS: DATA

START:
    MOV AX, DATA
    MOV DS, AX

LOOP_START:
    LEA DX, MSG1
    MOV AH, 9
    INT 21H

    MOV AH, 1
    INT 21H

    CMP AL, '0'
    JB NOT_NUM
```

```
30
31      CMP AL, '9'
32      JA NOT_NUM
33
34      MOV ANS, AL
35
36      LEA DX, MSG3
37      MOV AH, 9
38      INT 21H
39
40      MOV DL, ANS
41      MOV AH, 2
42      INT 21H
43
44      JMP LOOP_CHECK
45
46  NOT_NUM:
47      CMP AL, 'A'
48      JB ILLEGAL
49
50      CMP AL, 'F'
51      JA ILLEGAL
52
53      MOV ANS, AL
54      SUB ANS, 11H
55
56      LEA DX, MSG4
57      MOV AH, 9
58      INT 21H
59
60      MOV DL, ANS
61
62      MOV AH, 2
63      INT 21H
64      JMP LOOP_CHECK
65
66  ILLEGAL:
67      LEA DX, MSG5
68      MOV AH, 9
69      INT 21H
70
71  LOOP_CHECK:
72      LEA DX, MSG2
73      MOV AH, 9
74      INT 21H
75
76      MOV AH, 1
77      INT 21H
78
79      CMP AL, 'y'
80      JZ LOOP_START
81
82      CMP AL, 'Y'
83      JZ LOOP_START
84
85      MOV AH, 4CH
86      INT 21H
87
88  CODE ENDS
89  END START
```

Output

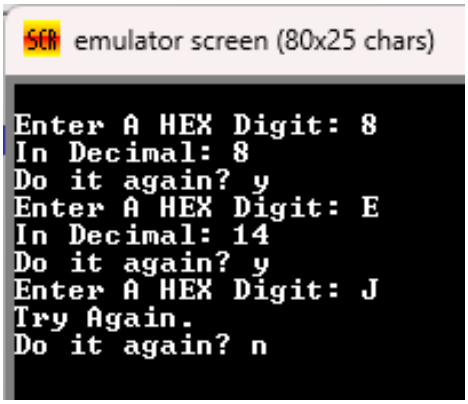


Figure 3: Converting HEX to Decimal

Problem 4.4

Problem Description

Write a program to read one of the HEX digits A-F, and display it on the next line in decimal.

Process

Code:

```
—

DATA SEGMENT
    CR EQU 0DH
    LF EQU 0AH
    MSG1 DB CR, LF, "Enter the HEX number : $"
    MSG2 DB CR, LF, "In decimal: 1$"
    NUM DB ?
    DEC_NUM DB ?
DATA ENDS

CODE SEGMENT
    ASSUME CS:CODE, DS DATA

START:
    MOV AX, DATA
    MOV DS, AX

    LEA DX, MSG1
    MOV AH, 9
    INT 21H

    MOV AH, 1
    INT 21H
    MOV NUM, AL

    SUB NUM, 11H

    LEA DX, MSG2
    MOV AH, 9
    INT 21H

    MOV DL, NUM

    MOV AH, 02H
    INT 21H
```

```
36      MOV AH, 4CH
37      INT 21H
38  CODE ENDS
39  END START
```

Output

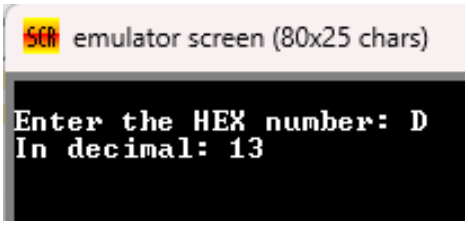


Figure 4: Converting HEX (A-F) to Decimal

Problem 4.5

Problem Description

Process

Code:

```
—

DATA SEGMENT
    CR EQU 0DH
    LF EQU 0AH
    STARS DB CR, LF, "**********$"
    COUNTER DW 10
DATA ENDS

CODE SEGMENT
    ASSUME CS:CODE, DS DATA

START:
    MOV AX, DATA
    MOV DS, AX
    MOV CX, COUNTER

LOOP_START:
    LEA DX, STARS
    MOV AH, 9
    INT 21H

    DEC CX
    CMP CX, 0

    JE LOOP_END
    JMP LOOP_START

LOOP_END:
    MOV AH, 4CH
    INT 21H

CODE ENDS
END START
```

Output

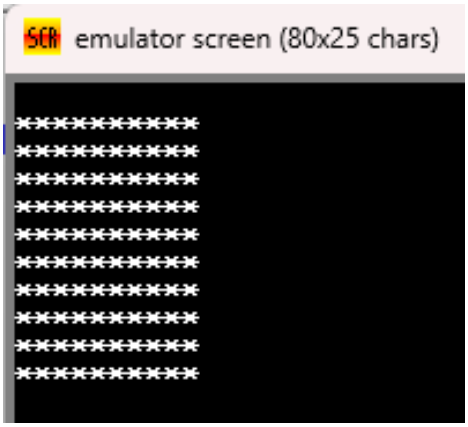


Figure 5: Printing a pattern using Loop

Keywords

These are the new keywords learned while solving the Problems

CMP Used to compare 2 provided byte/word.

JMP Used to jump to the provided address to proceed to the next instruction.

JA Used to jump if above/not below/equal instruction satisfies.

JB Used to jump if below/equal/ not above instruction satisfies.

JG Used to jump if greater/not less than/equal instruction satisfies.

JLE Used to jump if less than/equal/if not greater than instruction satisfies.

JZ Used to jump if equal/zero flag ZF = 1